Problem Set No. 1

Exercise 1. **I).** In a group of twenty people, is it possible that seven of them each have exactly three friends, nine of them have exactly four, and four of them have exactly five? Justify.

II). **Mr** and **Mrs X** attend a meeting. There are three other couples in the audience and several handshakes are exchanged. Nobody shakes their own hand and spouses do not shake hands with each other. Any two people in the gathering shake hands at most once.

Mr X notes that the seven other people exchanged handshakes in numbers that are all distinct.

1. Using a graph, state the possible values for the number of handshakes made by each person and deduce the value for **Mr X**.

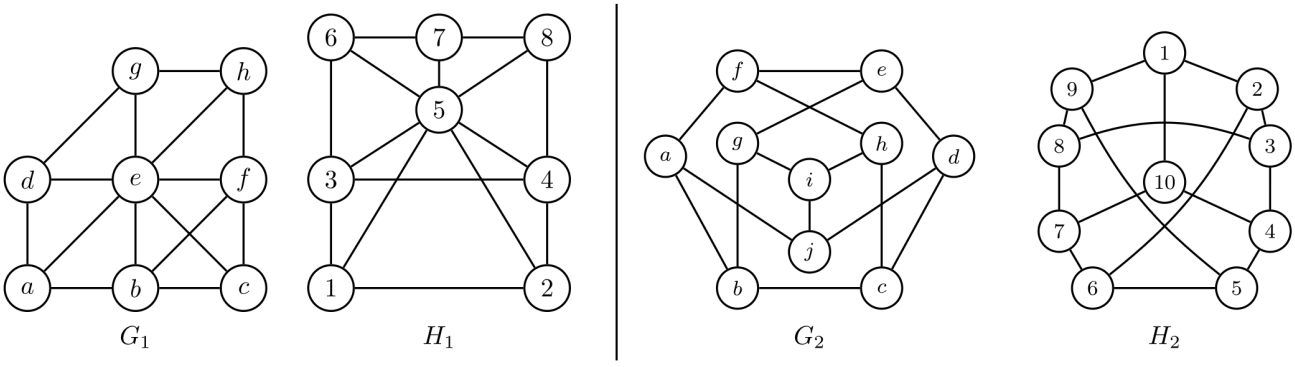
III). For each of the sequences below, decide whether it is the list of vertex degrees of a simple graph.

1. 7, 6, 5, 4, 3, 2, 1,
2. 6, 6, 5, 4, 3, 3, 1,
3. 3, 3, 2, 1, 1,

Exercise 2. Let G be a graph of order n and size m. Show that:

1. $m \leq \dfrac{n(n-1)}{2}$ and $\delta \leq \dfrac{2m}{n} \leq \Delta$;
2. If G is bipartite then $m \leq \dfrac{n^2}{4}$
3. If G is regular and n is odd then G is not bipartite;
4. The graph G contains a pair of vertices of the same degree;
5. If G has exactly two vertices of odd degree then there exists a chain linking them.

Exercise 3. **I.** For each case, say, with justification, whether the graphs G_i and H_i, for $i = 1, 2$, are isomorphic:



G_1 H_1 G_2 H_2

II. Consider G and H two graphs where S is the set of vertices of degree r in G and T the set of vertices of degree r in H

1. Show that if G and H are isomorphic then the induced subgraphs $G(S)$ and $H(T)$ are isomorphic.
2. Give an example of two non-isomorphic graphs having the same order, the same size and the same degree sequence but for which $G(S)$ and $H(T)$ are not isomorphic.

Exercise 4. An n-cube Q_n (or hypercube of dimension n ($n \geq 1$)) is a graph whose vertices represent the elements of $\{0,1\}^n$, and where two vertices are adjacent if and only if the corresponding n-tuples differ in exactly one component.

1. Draw Q_1; Q_2; Q_3 and determine the order of Q_n; the degree of each vertex of Q_n and then deduce the size of Q_n;
2. Show that Q_n is bipartite;
3. Say for which values of n the hypercube is Eulerian;
4. Write Q_n in terms of the Cartesian product of graphs and deduce that Q_n is Hamiltonian for all $n \geq 2$;
5. Determine the distance between any two vertices of Q_n using the binary vectors representing x and y and deduce the diameter of Q_n.

Exercise 5. Let G be a connected graph of order $n \geq 3$. Suppose that every vertex is coloured with one of the colours: red, blue or green such that for each given colour i, there exists at least one vertex x of G of colour i:

1. Show that G contains two adjacent vertices coloured differently.
2. Show that, independently of the order of G, G may contain no adjacent vertices with the same colour;
3. Show that there exists a chain P containing at least one vertex of each of the three colours.

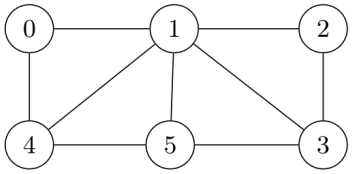
Exercise 6. Let $G = (V, E)$ be a graph of order n and size m and consider $L(G)$ the edge-representing graph (adjoint graph, line graph):

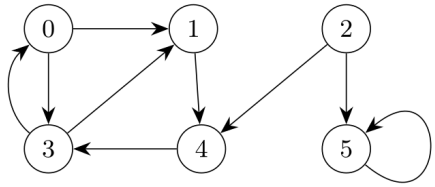
1. Show that $L(G)$ has m vertices and $\sum_{x \in V} \binom{2}{d(x)}$ edges;
2. Show that if G is Eulerian then $L(G)$ is Hamiltonian and Eulerian;
3. Show that the complement of the line graph of K_5 is isomorphic to the Petersen graph $P(5, 2)$.

Exercise 7.

1. Show that the complete multipartite graph $K_{n_1, \ldots, n_k}$ of order $n = n_1 + \cdots + n_k$ has size $\frac{1}{2}\left(n^2 - \sum_{i=1}^{k} n_i^2\right)$. Check on $K_{3,2,2}$.
2. Determine the complement of $K_{n_1, \ldots, n_k}$.
3. The Petersen graph is 3-regular of order 10. What is its size? What are the order, the size and the regularity of its complement?
4. Show that $K_{p,q}$ is regular if and only if $p = q$. Deduce the order and size of an r-regular complete bipartite graph.
5. Show that a bipartite graph of order n has at most $\lfloor n^2/4 \rfloor$ edges. Which graphs achieve equality?
6. Show that a forest of order n with c connected components has $n - c$ edges (we admit that a tree of order p has $p - 1$ edges).

Exercise 8. Determine the incidence matrix, the adjacency matrix and the adjacency list of the following graphs:





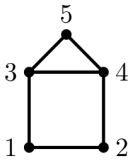
Exercise 9. I) Draw the graph corresponding to the following adjacency matrix:

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

II) Are the following matrices the incidence matrices of a simple undirected graph? If so, draw this graph, otherwise give the reason:

$$M_1 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

III) Complete the following table:

Set definition	Graphical representation	Adjacency matrix	Incidence matrix	Adjacency list
				
$V = \{1, 2, 3, 4, 5\}$ $E = \{12, 45, 34, 14, 35\}$				
		$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$		
			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	
				— 1 : [2, 3, 5] — 2 : [1, 4] — 3 : [1, 5] — 4 : [2, 5] — 5 : [1, 3, 4]

Exercise 10. A graph on 7 vertices is given by its adjacency matrix M below. This graph represents the 7 benches of a park and the paths that allow going from one to the other.

$$M = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

1. We want to paint the benches so that two benches linked by a path always have different colours. Give bounds on the minimum number of colours needed, with justification. Determine this number.
2. Is it possible to walk along all the paths of this park without going along the same path twice?
3. Is it possible to walk along paths of this park passing by each bench exactly once?
4. Compute M^2 and M^3. What do the numbers obtained correspond to?

Exercise 11. Let $H = (V, E)$ be hypergraph (b) of the course: $V = \{a, \ \dots, h\}$ and

$$e_1 = \{a, b, c\},\ e_2 = \{b, d\},\ e_3 = \{b, c\},\ e_4 = \{c, d, e\},$$
$$e_5 = \{c, e\},\ e_6 = \{e, f, g\},\ e_7 = \{f, g\},\ e_8 = \{f, h\}.$$

1. Write the vertex-hyperedge incidence matrix M ($M_{ij} = 1$ if $v_i \in e_j$). What do the sums of its rows represent? Of its columns? What does M become for a graph?
2. The **2-section graph** of H has vertex set V; two vertices are adjacent in it if they belong to a common hyperedge. Determine its edges, its clique number ω and its stability number α.
3. A **transversal** of H is a set $T \subseteq V$ that meets every hyperedge; a **stable set** of H is a set $S \subseteq V$ containing no hyperedge. Show that S is a stable set of H if and only if $V \setminus S$ is a transversal of H. Determine the minimum cardinality $\tau(H)$ of a transversal (three pairwise disjoint hyperedges give a lower bound), then the maximum cardinality $\alpha(H)$ of a stable set of H.
4. Compare with property 1 of the course when all hyperedges have cardinality 2.